QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION OCTOBER 2026
FIFTH SEMESTER
DSE – MG5DSEMBG300 – MEDICAL BACTERIOLOGY I
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Name the dye used as substrate/artificial electron donor for oxidase test. [K] / [CO 1]
2. Write any one selective medium used for *Staphylococcus aureus*. [K] / [CO 2]
3. Name the neurotoxin toxin responsible for tetanus. [K] / [CO 3]
4. Recall the serological test used for diagnosing Enteric fever. [K] / [CO 4]
5. Identify the etiology of meningococcemia. [A] / [CO 2]

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

6. Explain the interpretation of a TSI result with alkaline slant and acidic butt. [U] / [CO 1]
7. Determine the application of tube coagulase test [A] / [CO 2]
8. List the major toxins of *Clostridium perfringens* and their significance. [K] / [CO 3]
9. What is Quellung reaction? [K] / [CO 2]

10. Explain the use of toxoid with an example. [A] / [CO 3]

11. What is swarming? [K] / [CO 1]

12. Explain significant bacteriuria. [U] / [CO 4]

13. Explain the biochemical characterization of Klebsiella [A] / [CO 4]

14. Illustrate the difference in morphology of *Neisseria* spp. [U] / [CO 2]

15. Explain Lancefield groups. [U] / [CO 2]

16. What is infant botulism? [K] / [CO 3]

17. Analyze the role of *Pseudomonas aeruginosa* in nosocomial infections. [An] / [CO 4]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Explain the principle and interpretation of MR and VP test. [U] / [CO 1]

19. Describe the morphology and cultural characteristics of *Sterptococcus pneumoniae* [K] / [CO 2]

20. Explain the pathogenesis of non-suppurative Streptococcal diseases. [A] / [CO 2]

21. Summarize the laboratory diagnosis of diphtheria. [U] / [CO 3]

22. Explain the etiology and clinical features of human anthrax. [U] / [CO 3]

23. Classify Vibrios. Explain the serological classification introduced by Gardner and Venkatraman. [An] / [CO 4]

24. Categorize the pathogenesis of Diarrheagenic *E. coli*. [An] / [CO 4]

[6x5 = 30]

Part D

Essay Questions

Answer any **1** questions

Each question carries **15** marks

25. Describe the virulence factors and pathogenesis of Staphylococcal infections and intoxications. [K] / [CO 2]

26. Explain the etiology, pathogenesis and laboratory diagnosis of bacillary dysentery [U] / [CO 4]

[15x1 = 15]